# Ericsson PE Node ELEC-PE-06 Site Electronic City Hub QoS/Congestion Event Resolution SOP

**Node:** ELEC-PE-06 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_9293

## Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a QoS/Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure on the Ericsson PE Node ELEC-PE-PE-06 located at the Electronic City Hub. The underlying cause of this failure domain is attributed to the vendor-specific implementation of the QoS policy on the PE node, which may lead to microbursts and interface tail drops due to misconfigured traffic policing or shaping. Additionally, the failure may be exacerbated by hardware limitations such as insufficient buffering capacity or interface congestion. The PE node's role in the network architecture, coupled with the Ericsson vendor-specific CLI dialect, necessitates a tailored approach to diagnose and resolve this failure domain. This SOP provides a step-by-step guide to isolate the root cause, implement corrective actions, and verify the resolution.

## Pre-requisite Checks

- Ensure the Electronic City Hub is accessible for remote hands to perform maintenance
- Verify the redundant link status to ensure network redundancy
- Check the node's logging and monitoring systems for recent events and errors

## Potential Root Causes

| Cause | Probability |
|---|---|
| Misconfigured QoS policy leading to microbursts and interface tail drops | High |
| Insufficient buffering capacity on the PE node | Medium |
| Interface congestion due to high traffic volume | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show police, Expected Output: Police configuration for interface [interface name], Description: This command displays the current police configuration for the interface, allowing the operator to verify if the QoS policy is correctly configured.

Command: show shape, Expected Output: Shaping configuration for interface [interface name], Description: This command displays the current shaping configuration for the interface, enabling the operator to assess if the traffic is being correctly policed or shaped.

Command: show interface, Expected Output: Interface statistics, including errors and drops, Description: This command provides a comprehensive view of the interface statistics, including errors and
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```
drops, which can help identify potential congestion issues.
```

## Resolution Steps

Step 1: Step: Reconfigure QoS policy to prevent microbursts and interface tail drops, Command: config police [interface name] [new police configuration], Description: The operator should reconfigure the QoS policy to prevent microbursts and interface tail drops by adjusting the police configuration as needed.

Step 2: Step: Increase buffering capacity on the PE node, Command: config buffer [interface name] [new buffer size], Description: If the root cause is attributed to insufficient buffering capacity, the operator should increase the buffer size on the PE node to alleviate congestion.

Step 3: Step: Implement traffic management measures to prevent interface congestion, Command: config traffic-management [interface name] [new traffic management configuration], Description: To mitigate interface congestion, the operator should implement traffic management measures, such as traffic shaping or policing, to regulate traffic flow.

## Post-Check Verification

```
Command: show police, Expected Output: Police configuration for interface [interface name] (reconfigured), Description: After implementing corrective actions, the operator should verify that the QoS policy has been correctly reconfigured to prevent microbursts and interface tail drops.

Command: show shape, Expected Output: Shaping configuration for interface [interface name] (reconfigured), Description: The operator should also verify that the shaping configuration has been correctly reconfigured to prevent interface congestion.

Command: show interface, Expected Output: Interface statistics, including errors and drops (reduced), Description: Finally, the operator should verify that the interface statistics, including errors and drops, have been reduced to confirm the resolution of the issue.
```

## Roll-back Procedures

- Step: Revert QoS policy configuration to previous state, Command: config police [interface name] [previous police configuration], Description: In case the resolution fails, the operator should revert the QoS policy configuration to its previous state to restore network functionality.

- Step: Revert buffer size configuration to previous state, Command: config buffer [interface name] [previous buffer size], Description: Similarly, if the resolution fails, the operator should revert the buffer size configuration to its previous state to restore network functionality.

- Step: Revert traffic management configuration to previous state, Command: config traffic-management [interface name] [previous traffic management configuration], Description: Finally, if the resolution fails, the operator should revert the traffic management configuration to its previous state to restore network functionality.